Systematic Comparison of Multiple Retrieval-Augmented Generative AI Architectures for Evidence-Based Medical Question Answering with Explainability and Hallucination Control

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Abstract

LLMs are now widely used in healthcare to assist in clinical tasks, but they still produce answers that sound correct but are totally made up. In the medical field, these false outputs may result in wrong diagnoses, harmful treatments and a severe risk to patients. RAG has been introduced to reduce this issue by pulling in real evidence from trusted sources before generating an answer but different RAG designs retrieve evidence in different ways. Despite this, very limited research was done comparing multiple RAG architectures side by side on the same medical dataset using the same language model in controlled conditions. This research compares four RAG architectures, Naive RAG, Sparse RAG, Hybrid RAG and Multi-Hop Explainable RAG on the MedQA USMLE dataset containing 12,723 clinical questions. All four systems use the same LLM, embedding model, prompt template and retrieval setup, so all that is changing between experiments is how evidence is retrieved. Each architecture is evaluated at three layers: the answer accuracy, hallucinations detection and retrieval quality in terms of using five RAGAS metrics and explainability in terms of using LIME and SHAP to check which evidence passages influenced which answers. The expected outcome is a practical comparison framework which reveals which RAG design produces the most accurate and least hallucinated, most explainable medical answers. This research is designed to offer concrete and evidence-based recommendations when it comes to selecting the appropriate RAG architecture when constructing dependable AI systems for healthcare.

List of Abbreviations

AI: Artificial Intelligence

LLM: Large Language Model

RAG: Retrieval-Augmented Generation

AMA: American Medical Association

QA: Question Answering

RAGAS: Retrieval Augmented Generation Assessment

LIME: Local Interpretable Model-agnostic Explanations

SHAP: SHapley Additive exPlanations

MedQA: Medical Question Answering

USMLE: United States Medical Licensing Examination

XAI: Explainable Artificial Intelligence

API: Application Programming Interface

BM25: Best Matching 25

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CoT: Chain-of-Thought

CUDA: Compute Unified Device Architecture

FAISS: Facebook AI Similarity Search

F1: F1 Score

FN: False Negative

FP: False Positive

GPU: Graphics Processing Unit

IDF: Inverse Document Frequency

JSONL: JSON Lines

NLP: Natural Language Processing

OMML: Office Math Markup Language

RRF: Reciprocal Rank Fusion

SMART: Specific, Measurable, Achievable, Relevant, Time-bound

SSD: Solid State Drive

TP: True Positive

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1. Background

In cases where a LLM with certainty informs a doctor that drug interaction is safe but, in turn, the news turns out to be false, the results can be life-threatening. In an oncology study, it was recently demonstrated that about two out of five medical responses generated by AI had an incorrect or unsubstantiated statement (Mi Yoon et al., 2025). These are the mistakes also known as hallucinations in which a model creates a piece of text that seems convincing, but is either historically inaccurate or lacks support based on verifiable facts. Such errors could go unnoticed in normal applications, but in the healthcare field directly result in misdiagnosis, wrong treatment or delayed intervention. It has been confirmed that several studies have not yet performed enough to verify the reliability and transparency of the use of LLMs (Thirunavukarasu et al., 2023)

Nevertheless, these fears do not mean that the use of AI in healthcare has decreased. LLMs have been adopted in real hospitals and clinics to aid in doing such things as summarising patient records, help in diagnosis and answering medical questions. AMA revealed that the number of physicians using AI based tools increased as of 2023 by 38 percent but by 2024, the figure had increased to 66 percent (AMA, 2025) which demonstrated that AI is no longer in the experimental phase of application but has become an actual component of clinical practise. This fast adoption further intensifies the rapidity of the need to resolve the problem of hallucinations before these systems cause real harm.

Researchers came up with a method to solve this issue, referred to as RAG (Lewis et al., 2021). The general concept is straightforward as opposed to having to entirely rely on what a model was trained to do, the system will at first access a relevant information in trusted external sources like medical journals, textbooks or clinical databases and then apply the evidence that it has retrieved to create its answer. This step of grounding is useful in minimising the possibility of the model going off on its own.

RAG systems do not behave just like that. Naive RAG uses a simple retrieve then generate pipeline, which loads a fixed number of documents to each query without paying attention to query complexity. Sparse RAG only retrieves the outside knowledge in the case of a low internal confidence of the model and does not need extra retrieval (Asai et al., 2023). Hybrid RAG uses sparse lexical similarities such as BM25 with dense neural embeddings to discover both literal similarities of keywords and semantic similarities (Karpukhin et al., 2020;(Sawarkar et al., 2024). Multi-Hop Explainable RAG is a multi-round explainable retrieval method that utilises a cumulative approach to attach disparate evidence across sources to respond to complex clinical queries (Trivedi et al., 2023).

Though these developments take place, few studies have focused on comparison of these architectures based on the same set conditions. Available literature is inclined to measure retrieval accuracy and answer correctness separately, but it does not consider hallucination behaviour and explainability when using controlled experimental conditions. Majority of the evaluations only evaluate a single combination of RAGs without comparing the two or more architectures on the same medical dataset (Miao et al., 2025). It implies that it is not clear which architecture is the best when the only variable that is being manipulated is retrieval strategy in medical QA.

A comparative analysis of four RAG architectures can be proposed, namely Naive RAG, Sparse RAG, Hybrid RAG and Multi-Hop Explainable RAG using the MedQA data set as a standardised benchmark to address this issue. All of the experiments will use the same underlying language model and dataset with only the retrieval architecture different. The test will not rely merely on the accuracy but also will assess the decrease in hallucination, retrieval relevance and explainability. This way, this study will determine the best and most reliable RAG architecture upon which reliable medical AI systems can be built and offer useful advice.

2. Problem Statement and Literature Review

2.1 Problem Statement

LLMs are increasingly being used in healthcare to answer medical questions, but two fundamental problems prevent their safe adoption in clinical settings, hallucination and lack of explainability. When these models generate answers there is no reliable way to verify whether the response is grounded in real medical evidence or fabricated by the model itself. In a clinical environment a single hallucinated drug dosage or an incorrect diagnosis can lead to wrong treatment decisions, delayed interventions or in the worst case lead to preventable patient death. Medical professionals rely on accurate information to make life or death decisions and when an AI system confidently presents false information as fact its consequences are far more severe than in any other domain (Ji et al., 2024).

The introduction of RAG was done to lessen hallucination by basing model responses on externally retrieved evidence. Although this is better than standalone language models, it also very depends on retrieval design. When the retrieval step cannot retrieve the appropriate evidence, the language model attempts to fill in the blanks independently by generating answers which sound medically convincing but have no support whatsoever. Various architectures do evidence collection fundamentally differently but there is very little controlled comparison, which compares these methods on the same medical dataset under the same language model (Xiong et al., 2024). In the absence of such comparison, there is no clear indication that the healthcare organizations should take into consideration when selecting the most credible RAG architecture to be used in the medical field. This study fills that gap.

Key Problem Areas Identified in Literature:

- Existing medical RAG studies mostly evaluate individual, making it difficult to isolate the effect of retrieval strategy alone.(Xiong et al., n.d.)(Miao et al., 2025)

- Hallucination behavior across different RAG architectures has not been systematically measured, meaning a system could be silently hallucinating without anyone knowing (Tonmoy et al., 2024).
- Explainability remains largely absent from RAG evaluations, with most studies neglecting whether answers can be traced back to specific evidence passages (Chaddad et al., 2023).
- A single evaluation covering accuracy, hallucination behavior and explainability across multiple RAG architectures remains largely unexplored in the medical domain.

2.2 Literature Review

Category 1: RAG in Healthcare

Studies in this category apply RAG techniques specifically within healthcare and medical domains. The reviewed works demonstrate that grounding language model responses in external medical knowledge improves answer accuracy and reduces hallucination but they also reveal significant gaps in architectural comparison and explainability evaluation.

Table 2.1: Literature Review, Category 1 - RAG in Healthcare

Summary	Research Gaps
(Xiong et al., 2024)introduce benchmarking frameworks for medical RAG systems and show that using retrieval improves answer accuracy and reduces hallucinations compared to using language models alone. By testing different combinations of retrievers, medical knowledge sources, and language models on datasets such as MedQA-US, the study shows that choosing the right data source matters. It also finds that combining multiple retrieval rankings can further improve performance.	The study mainly evaluates basic or simple RAG setups and does not experiment with more advanced RAG architectures such as hybrid or multi-hop systems. In addition, the evaluation focuses mostly on accuracy and hallucination reduction and does not deeply analyses explainability features or confidence measures, which are important for building trustworthy medical AI systems.
(Miao et al., 2025) present a scoping review that analyses many RAG studies in medicine and nursing and shows a shift towards more logic-driven and active retrieval systems. It highlights the growing use of hybrid retrieval and query decomposition to reduce hallucinations and improve context handling.	The review points out that most existing RAG systems still lack strong reasoning capabilities. While it categorizes different types of RAG architectures, it does not experimentally compare their performance on the same dataset, leaving it unclear which approach works best in practice for medical QA.
(Chen et al., 2021) propose a multi-hop question answering framework that extracts clear reasoning chains, allowing humans to see and verify how an answer was derived. This improves both performance and interpretability on complex question answering tasks.	The approach focuses on extractive or multiple-choice answers rather than generating full natural language responses using modern language models. It is also tested on general-domain datasets, and adapting the method to medical terminology and clinical entities remains an open research area.

(Wampler et al., 2025)present a systematic review that organises different RAG architectures into a unified framework and highlights that graph-based and agentic RAG systems are more suitable for complex domains like medicine. It also introduces a framework to analyse trust and risk in RAG systems.	While the review discusses many RAG variants, it does not experimentally compare them on medical benchmarks such as MedQA. It also focuses more on citation-based trust than on generating clear, human-readable explanations of the reasoning process.
(Ke et al., 2024)build a medical RAG system using a set of preoperative clinical guidelines to support surgical fitness assessments. The system is tested on a small set of clinical scenarios and shows that a RAG-based approach can achieve very high accuracy, even outperforming human clinicians and base language models.	The study tests only a single, basic retrieval strategy and does not compare different RAG architectures such as naive, hybrid, or multi-hop approaches. It also focuses mainly on answer accuracy and does not include clear quantitative measures for explainability or the quality of cited evidence.

Category 2: RAG Architecture Evaluation

Studies in this category evaluate and compare different RAG architectures across various domains. The reviewed works provide evidence that architectural design choices significantly affect retrieval quality and answer accuracy but most comparisons are conducted outside the medical domain or lack controlled experimental setup with a shared baseline model.

Table 2.2: Literature Review, Category 2 - RAG Architecture Evaluation

Summary	Research Gaps
(Ersoy & Ersahin, 2025) compare sparse, dense, hybrid and fusion RAG approaches, finding that fusion-based RAG achieves the best accuracy while hybrid RAG offers a strong balance between performance and speed.	The experiments use finance and e-commerce datasets rather than medical data, so the findings are not validated for healthcare. Multi-hop reasoning chains, which are important for complex clinical questions, are also not tested.
(Hammane et al., 2024) combine biomedical literature retrieval with a multi-agent self-evaluation loop, achieving strong results on biomedical QA but only moderate performance on USMLE-style medical exam questions.	The system does not clearly outperform simpler RAG approaches on MedQA-style questions, raising doubts about whether the added complexity is justified. Controlled comparisons with simpler RAG baselines using the same language model are also missing.
(Zhao et al., 2025) propose a medical RAG system that uses a diagnostic knowledge graph to better distinguish between diseases with similar symptoms. The system breaks down patient information into clinical features and asks follow-up questions when important details are missing, leading to more accurate diagnoses.	The system is mainly tested on patient record style datasets focused on diagnosis, whereas MedQA contains more theory-based, multi-step reasoning questions. The approach relies on knowledge graphs built from clinical records, and adapting this method to textbook-based medical knowledge is still an open challenge.

(Al Ghadban et al., 2023) develop a medical RAG system to support community health workers by indexing clinical guidelines and evaluating different retrieval and generation configurations. The study shows that RAG improves traceability and explainability in medical education and finds a practical balance between retrieval quality and system speed.	The system is tested mainly on simple guideline lookup questions and does not evaluate performance on complex, multi-step clinical reasoning tasks like those in MedQA. More advanced retrieval architectures are avoided due to performance concerns, leaving open questions about their necessity and benefits for harder medical QA problems.
(Peng et al., 2023) discuss the limitations of current LLMs for evidence-based medicine, highlighting issues such as outdated knowledge, factual errors, and lack of explainable reasoning. The work argues that LLMs should be combined with structured medical knowledge sources to support trustworthy and explainable clinical decision-making.	The paper calls for benchmarks and evaluation metrics for medical reasoning but does not provide a concrete system or experimental framework. It also suggests combining language models with knowledge bases but does not implement such an architecture, leaving an implementation gap that this thesis aims to address.
(Khan et al., 2025) apply RAG to explain engineering exam solutions and show that retrieval-based grounding improves the faithfulness and relevance of generated explanations. The study highlights the importance of choosing the right model and embedding combination for technical question answering tasks.	The study keeps the overall RAG architecture fixed and mainly tunes model components, leaving open questions about how different RAG architectures themselves would perform. In addition, engineering exam questions are structurally different from medical questions, which often require multi-step clinical reasoning.

Category 3: Hallucination Reduction in Large Language Models

Research in this category focuses on detecting and reducing hallucinations in language model outputs. The reviewed works establish that retrieval based grounding and verification mechanism are effective strategies for reducing fabricated content but none provide a direct comparison of multiple RAG architectures specifically on medical question answering benchmarks.

Table 2.3: Literature Review, Category 3 - Hallucination Reduction

Summary	Research Gaps
(Tonmoy et al., 2024) survey hallucination reduction techniques including retrieval-based methods, self-checking mechanisms and knowledge graph grounding, highlighting that external grounding and verification steps are especially important for safety-critical fields like medicine.	The paper does not directly compare different RAG architectures for medical question answering or evaluate hallucination behaviour on medical QA benchmarks, leaving a gap in domain-specific analysis.
(Andriopoulos & Pouwelse, 2023) survey retrieval-based, graph-based and search-based methods for augmenting language models with external knowledge, emphasising that combining internal model knowledge with external sources improves grounding and scalability.	The survey notes that the interaction between reasoning and knowledge integration remains underexplored, and does not address how graph-based methods can produce natural language explanations in an explainable RAG setting.

(Hossain et al., 2025) survey explainable AI methods for medical data and categorise them into different explanation techniques for images, text, and tabular data. The work highlights that current explanation methods often fail robustness tests in clinical settings and calls for better human-AI interaction and evaluation standards.	The review discusses explainability in general but does not specifically frame retrieval-based grounding as an explainability mechanism, where retrieved sources themselves act as explanations. It also does not deeply address hallucination-specific evaluation metrics, which are central to this thesis.
(Patrice Béchard & Marquez, 2024) present a commercial RAG system that uses retrieval to ground code generation and significantly reduce hallucinated outputs in structured workflows. The results show that RAG can reduce hallucinations even when using smaller language models, making it a resource-efficient approach.	The study validates hallucination reduction in structured, non-medical domains and does not address factual correctness of medical statements. It also retrieves structured objects rather than unstructured medical text, leaving open questions about how retrieval granularity and document chunking strategies affect hallucination control in medical QA.

Category 4: Multi-Hop Reasoning in Natural Language Processing

Studies in this category explore multi-hop reasoning techniques, where systems perform multiple steps of evidence retrieval and reasoning to answer complex questions. The reviewed works demonstrate that structured, iterative reasoning paths improve performance on complex tasks but most are tested on general domain datasets and do not integrate with generative RAG system for medical applications.

Table 2.4: Literature Review, Category 4 - Multi-Hop Reasoning

Summary	Research Gaps
(Yu et al., 2023) propose a framework for generating multi-hop questions using entity graphs and reasoning chains extracted from text. Their generated questions improve downstream QA performance on general-domain datasets.	The work focuses on question generation rather than evidence-based answering, and its reasoning structures are tested on general text, not clinical data, so their relevance to medical QA remains unproven.
(Huo & Zhao, 2020) propose a sentence-level multi-hop reasoning model that selects linked sentences instead of full documents, reducing retrieval noise and improving performance on complex reasoning tasks.	The model is built for discriminative question answering and does not generate full natural language responses, while this thesis focuses on generative RAG systems. It also uses older embedding methods, so its performance with modern dense embeddings remains unclear.
(Wang et al., 2024)present a hierarchical reinforcement learning framework for multi-hop reasoning over knowledge graphs, producing interpretable reasoning paths and strong performance on structured reasoning tasks. The model demonstrates that structured multi-step reasoning can be learned effectively on graph-based data.	The approach focuses on identifying target entities in structured graphs and does not integrate the reasoning paths into a generative language model to produce full medical answers. It is also designed for structured knowledge graphs, whereas this thesis focuses on unstructured medical text.

(Sakarvadia, 2024) studies why multi-hop reasoning fails in language models and shows that explicitly injecting missing intermediate knowledge can greatly improve performance. The work also provides tools to visualise internal attention patterns to improve interpretability.	The approach relies on manually injecting missing knowledge into model internals, whereas this thesis focuses on automatically retrieving missing information through RAG. The work also evaluates short-form predictions rather than full natural language medical explanations.
(Amjad et al., 2023) review attention-based explainability methods used in healthcare NLP models and show that while attention mechanisms can highlight relevant text and achieve high predictive performance, the explanations are often limited to visual cues such as highlighted words.	The review focuses on implicit explanation methods based on model attention, whereas retrieval-based explanations provide explicit evidence sources. It also notes the lack of standard benchmarks for evaluating explanation quality, which this thesis addresses by using retrieval-based explainability and objective evaluation metrics.

3. Research Questions

1. What is the relative impact of retrieval architecture (Naive, Sparse, Hybrid, and Multi-Hop) on the factual accuracy and rate of hallucinations of RAG based medical question answering systems when tested on the same data with the same language model?
2. What is the most faithfulness and context relevance score of the retrieval architecture using the RAGAS evaluation framework on 12723 MedQA USMLE questions?
3. How far can explainability algorithms like LIME and SHAP generated medical responses be attributable to particular retrieved passages of evidence through various RAG architectures?
4. What is the interaction between complexity of retrieval strategies and reduction of hallucinations in medical question answering and does multi hop retrieval result in statistically lower hallucination rates compared to single-pass retrieval strategies?

4. Aim & Objectives of the Study

Aim: The purpose of this research is to compare four RAG architectures which is Naive, Sparse, Hybrid and Multi-Hop Explainable for medical question answering under controlled conditions where only the retrieval strategy varies while all other components remain constant. The study aims to determine which architecture delivers the most accurate, least hallucinated and most explainable responses on the MedQA USMLE benchmark.

Objectives:

1. To implement and validate four RAG architectures (Naive, Sparse, Hybrid, and Multi-Hop Explainable) using a shared language model, embedding model and retrieval infrastructure on the MedQA USMLE dataset.
2. To evaluate and compare the factual accuracy of all four architectures against verified correct answers across 12723 USMLE style clinical questions.

3. To measure hallucination behavior across all four architectures using RAGAS metrics including Faithfulness, Context Precision, Context Recall, Answer Relevancy and Answer Correctness.
4. To analyze the explainability of each architecture by applying Lime and Shap at the passage level to trace generated answers back to specific retrieved evidence.
5. To produce a comparison framework that maps clinical requirements such as accuracy, explainability and hallucination tolerance to the most suitable RAG architecture.

5. Significance of the Study

AI based systems are becoming more and more popular as medical professionals use them to aid in clinical decision-making, yet these systems remain vulnerable to generating false or unsupported information. One drug interaction or falsely made diagnosis can result in wrong medications and severe damages to the patient. As the proportion of physicians using AI tools has increased since 38% in 2023 to 66% in 2024, the hallucination addressing and explainability in medical AI is a patient safety concern and not just an academic exercise.

Academically, investigations that have been performed have assessed individual RAG architectures in isolation or applied them to general domain data that is not indicative of clinical reasoning complexity. There is very little research that compares several RAG architectures against each other on the same medical data with the same language model in a controlled way. This study offers a controlled comparison which is lacking in the field of research since it isolates the retrieval strategy as the only variable.

The explainability analysis, done with the help of LIME and SHAP, brings to the table a perspective that is currently absent in RAG assessments. The majority of studies are concentrated on accuracy and correctness, however, they do not investigate whether the correctness is possible to trace back to definite passages of evidence. In clinical practise, the correct answer that has no evidence trail would not suffice since the doctor must ensure that they make sense before they can act on the current reasoning.

In practical sense, the study results into a comparison paradigm that allocates clinical need to the most appropriate RAG architecture. Instead of suggesting one best system the framework enables organisations to choose an architecture with respect to their priorities of accuracy, hallucination risk or explainability. This comparison is reproducible and extendable due to use of five standardised RAGAS metrics.

Lastly, the study can help to ensure that medical AI systems are safer and more transparent. It offers actionable evidence on the success and failure of each RAG architecture, which makes it valuable in informed decision making by any person who constructs AI or deploys it in the healthcare field. Having detected hallucination and measured retrieval quality alongside explainable analysis in a single study would present a better picture than any of the individual evaluations could have presented.

6. **Scope of the Study**

Table 6.1: In-Scope & Out-Scope

In-Scope	Out-of-Scope
Implementation and evaluation of four RAG architectures - Naive RAG, Sparse RAG, Hybrid RAG and Multi-Hop Explainable RAG	Fine tuning or retraining language models on medical data
Use of the MedQA USMLE dataset (12,723 English multiple-choice clinical questions) as the evaluation benchmark	Testing on non-English medical datasets or clinical note datasets outside MedQA
Retrieval using FAISS for dense semantic search and BM25 for sparse keyword based retrieval	Use of knowledge graphs, graph based retrieval or agentic RAG systems
Evaluation using five RAGAS metrics: Faithfulness, Answer Correctness, Context Precision, Context Recall and Answer Relevancy	Real-time clinical deployment, patient-facing testing or integration with electronic health record systems
Explainability analysis at the passage level using LIME and SHAP across all four architectures	Development of new explainability methods beyond LIME and SHAP
Hallucination detection through RAGAS Faithfulness scoring and a three layer hallucination control strategy	Human expert evaluation by licensed medical professionals or clinical validation panels
Comparative analysis with statistical significance testing to identify performance differences across architectures	Comparison with non-RAG approaches such as standalone LLMs, prompt-only methods or supervised fine-tuned models

The scope is designed to directly support the aim and objectives of the study. Every in-scope item corresponds to a specific phase in the methodology, from architecture implementation through evaluation and explainability analysis. The out-of-scope boundaries prevent the research from expanding beyond what is achievable within the given timeline and available resources, while clearly identifying areas that future research can address. By defining these boundaries upfront, the study maintains focus on answering the four research questions through controlled and reproducible experiments.

7. Research Methodology

7.1 Overview of Methodology

The methodology section specifies how four RAG architectures are developed, realised and evaluated with respect to medical question answering, namely, Naive RAG, Sparse RAG, Hybrid RAG and Multi-Hop Explainable RAG. Each of the four architectures is trained on MedQA dataset with the same language model, embedding model, vector database and prompt template. The study can isolate what is actually caused by retrieval design on the quality of answers, the rates of hallucinations and explainability by leaving all other factors the same and solely altering the retrieval strategy.

The research pipeline passes through multiple related steps, namely dataset preparation, building a medical knowledge base by text chunking and embedding generation through implementing all four architectures and evaluating it by measuring both factual accuracy and detecting hallucinations using RAGAS metrics and explaining it using LIME and SHAP. One of the key issues here is related to the issue of hallucination control in which evidence-based prompting would limit the model to question contexts retrieved and faithfulness scoring that measures the extent to which a certain answer is supported by the evidence presented. According to the comparative results the research suggests a deployment-oriented framework that makes recommendations regarding the choice of the most appropriate RAG architecture of reliable medical AI systems.

7.2 Research Type and Strategy

7.2.1 Research Philosophy

The research philosophy is based on quantitative and experimental approach that is intended due to the primary objective of assessing the performance of the various AI systems design in a real environment where the domain of operation is high-risk such as medical question answering. Rather than concentrating on theoretical models alone, it supports its approach by developing working systems, trying them on realistic medical problems and evaluating their behaviour in terms of accuracy, reliability, explainability and the risk of hallucination.

7.2.2 Research Approach

A practical AI research model is engaged in which all the four RAG systems are created, tested and tested through formal experiments in controlled situations. The evaluation of quantitative data is done using the RAGAS framework of calculating accuracy, faithfulness, context precision and context recall as numeric measurements of all the test questions. Qualitative evaluation is concerned with the explainability, which is evaluated by the use of LIME and SHAP to consider whether each architecture can trace its responses based on certain bits of retrieved evidence. This two-fold strategy guarantees that not only the performance of each system is evaluated, but also the transparent manner in which it comes to its responses.

7.2.3 Research Design

An experimental, comparative and design-science methodology is taken. The experimental part will entail constructing and testing four RAG architectures in controlled conditions in which the only factor will be the retrieval strategy. The comparative element entails comparing all the four to one another with the same MedQA data, the same LLM, and the same RAGAS evaluation measurements by making sure that the difference in the performance due to any of the four can be linked to the retrieval architecture. The design-science aspect entails the process of developing and improving these systems through iterations following their performance as witnessed during validation experiments prior to the execution of the final evaluation.

Table 7.1: Research Design Summary

Component	Description
Research Type	Experimental, Comparative, Design-Science
Approach	Applied AI system development with structured experiments
Architectures	Naive RAG, Sparse RAG, Hybrid RAG, Multi-Hop Explainable RAG
Dataset	MedQA (USMLE-style, 12723 questions)
LLM	LLaMA 3.3 70B Versatile via Groq API (same across all architectures)
Quantitative Eval	RAGAS metrics: Faithfulness, Answer Correctness, Context Precision, Context Recall, Answer Relevancy
Qualitative Eval	Explainability analysis using LIME and SHAP

7.3 Dataset Selection and Preparation

7.3.1 MedQA Dataset Description

The MedQA dataset (Jin et al., 2020) is the first large-scale, open-domain medical question answering benchmark sourced from professional medical board examinations. It consists of 61,097 multiple-choice questions based on actual professional medical licencing tests, 12,723 of which are in English, 34,251 in simplified Chinese and 14,123 in traditional Chinese. In this study, we have been using English version of MedQA.

The data set is accompanied with a corpus of 18 English medical textbooks that will be used as the external knowledge base to be used in retrieval in all four RAG architectures. This will make sure that the retrieval component possesses access to rich verified medical literature as opposed to just using the knowledge of the language model.

Table 7.2: MedQA Dataset Structure

Field	Description	Example	Type
id	Unique identifier for each question	"usmle_001"	string
question	Clinical scenario or question stem	"A 45-year-old male presents with chest pain..."	string
opa	Option A	"Myocardial infarction"	string
opb	Option B	"Pulmonary embolism"	string
opc	Option C	"Aortic dissection"	string
opd	Option D	"Pneumonia"	string

correct	Index of correct answer (A=0, B=1, C=2, D=3)	1 (indicating Option B)	integer
subject	Medical subject	"Cardiology"	string

7.3.2 Preprocessing and Formatting of Data

The dataset undergoes a series of preprocessing stages before the execution of any experiment, and consistency and quality are maintained. As Figure 7.1 illustrates, the two data sources take different preprocessing channels. The MedQA data is loaded as a JSONL and filtered to remove duplicate questions and verified to make sure that each question has its option mappings are correct. The textbook corpus undergoes the process of the raw text extraction, medical text cleaning and normalisation of terminology. The two directions meet in text encoding standardisation and the cleaned text is then chunked and analysed into embeddings. The output is divided into two indices - FAISS vector index to dense semantic retrieval and BM25 keyword index to sparse. These two indices serve as the retrieval backends for all four RAG architectures in the study.

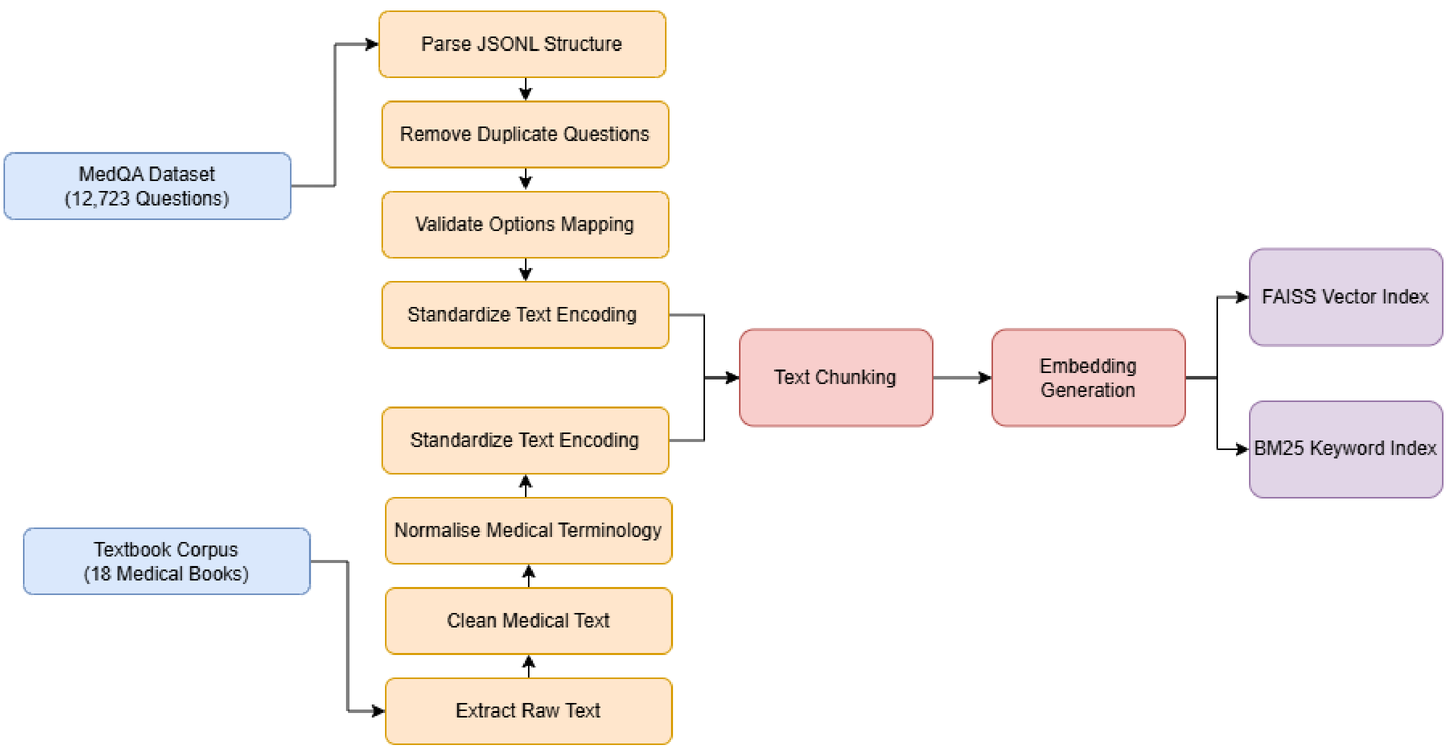

Figure 7.1: Data Preprocessing

7.4 Data Representation and Retrieval Preparation

This section shows how medical knowledge that is found in the textbook corpus will be transformed into a structured format that can be retrieved. It follows three major steps: text is done into smaller units, the units are turned into numerical representations and put into a vector database where they can be searched effectively.

7.4.1 Text Chunking Strategy

The medical textbook corpus must be subdivided into small manageable parts known as chunks before any retrieval can occur. The retrieval system is incapable of providing all the medical concepts relevant to a question and only the most relevant ones by dividing the text into focused units. The chunking strategies to be followed and evaluated to understand which method is best in medical question answering will be the following:

Table 7.3: Text Chunking Strategies

Strategy	Description	Justification of Use
Fixed-Size Chunking	Splits text into equal sized segments of a fixed token count, simple to implement but may cut across sentence or paragraph boundaries	Serves as a baseline to compare against more advanced strategies and measure whether simple splitting is sufficient for medical text
Overlapping Chunking	Adds a 10-20% overlap between consecutive chunks to preserve context that would otherwise be lost at chunk boundaries	Prevents loss of critical medical information that spans across chunk boundaries, such as drug interactions described across two paragraphs
Recursive Chunking	Splits text hierarchically using natural boundaries such as paragraphs, sentences and words, producing variable-length chunks that preserve logical structure	Selected as the primary strategy because medical textbooks are structured around topics, sections and paragraphs that carry complete clinical concepts
Semantic Chunking	Splits text where embedding similarity between consecutive sentences drops below a threshold, producing chunks that align with topic boundaries	Evaluated as an advanced alternative to test whether meaning-aware splitting improves retrieval relevance for complex medical questions

7.4.2 Embedding Model Selection and Implementation

Embedding models convert medical text into numerical vector representations that capture semantic meaning. These vectors enable similarity-based retrieval where the system finds chunks whose meaning is closest to the question being asked. The following embedding models have been shortlisted as candidates for this research, each offering different trade-offs between dimensionality, parameter count and medical domain performance. The final selection will be made based on retrieval relevance testing on the MedQA validation set before running the full experiments.

Table 7.4: Embedding Model Comparison

Model	Dimension	Params	Medical Performance
BGE-large-en-v1.5	1024	335M	80-84 nDCG@10 on TREC-COVID
MedCPT (NCBI)	768	110M	SOTA on 3/5 BEIR benchmark
all-MiniLM-L6-v2	384	22M	56% Top-5 Accuracy

7.4.3 Vector Database Implementation

A vector database is needed to store the chunk embeddings and enable fast similarity search during retrieval. Two popular options were considered for this research, FAISS and ChromaDB. After comparison, FAISS was selected as the primary vector database.

Table 7.5: Vector Database Comparison

Feature	FAISS	ChromaDB
Type	Library (in-memory)	Embedded database
Cost	Free (open-source)	Free (open-source)
Query Latency	0.34ms	2.58ms
GPU Support	Yes (CUDA)	No
Metadata Filtering	External only	Rich built-in
Reproducibility	Full (local)	Good (local)

FAISS (Facebook AI Similarity Search) is selected as the primary vector database for this research. It is the standard choice in academic RAG research and was used in the original RAG paper by (Lewis et al., 2021). Its key advantages include zero cost, deterministic behavior for reproducible experiments, GPU acceleration via CUDA and support for over 20 index types.

7.5 LLM Integration and Prompt Engineering

7.5.1 LLM Selection and Integration

A single LLM is used across all four RAG architectures to ensure fair comparison, so that performance differences are due to retrieval design rather than the model itself. The selected model is LLaMA 3.3 70B Versatile an open-weight model with 70 billion parameters accessed via the Groq API for fast inference. It supports a large context window of 131k tokens, enabling rich medical evidence to be provided as input and delivers high throughput, making large-scale experiments practical. The model shows strong performance on complex reasoning and medical knowledge tasks, comparable to leading models (Grattafiori et al., 2024).

7.5.2 Prompt Engineering Strategy

Various prompting techniques including zero-shot, few-shot, chain-of-thought, self-consistency and tree-of-thought will be evaluated during the validation phase to determine which approach best reduces hallucination and supports explainable evidence-grounded responses across all four RAG architectures.

Table 7.6: Prompt Engineering Techniques

Technique	Description	Justification for Use
Zero-Shot Prompting	The model is asked a biomedical question without prior examples or extra context	Serves as a baseline to evaluate LLM behavior and hallucination rates without any guidance
Few-Shot Prompting	A few biomedical Q&A examples are given to the model before the actual question	Tests whether exposure to relevant examples reduces unsupported outputs
Chain-of-Thought Prompting	Instructs the model to produce reasoning steps before answering	Encourages step-by-step logical reasoning, which can reduce factual errors
Self-Consistency Prompting	Model generates multiple reasoning paths and selects the most consistent answer	Filters out inconsistent or hallucinated responses through majority selection
Tree-of-Thought Prompting	Explores multiple reasoning branches and selects the best path before answering	Useful for complex medical queries that require structured diagnostic exploration
Instruction-Based with Evidence Grounding	Restricts the model to answer only from retrieved evidence and cite specific passages	Directly controls hallucination and enables explainability analysis through citation tracking

7.6 RAG Architecture Development and Retrieval Strategy Comparison

This section describes the implementation of four RAG architectures each using a different retrieval strategy but sharing the same embedding model, vector database, LLM and prompt template.

7.6.1 Naive RAG Architecture

Naive RAG uses a straightforward single-step retrieval approach based entirely on semantic similarity. As shown in Figure 7.2, the medical question is converted into an embedding vector which is used to retrieve the top-k most similar chunks from the FAISS index. The retrieved context is combined with the original question into a prompt and passed to the LLM for answer generation. This architecture serves as the baseline for the entire comparison.

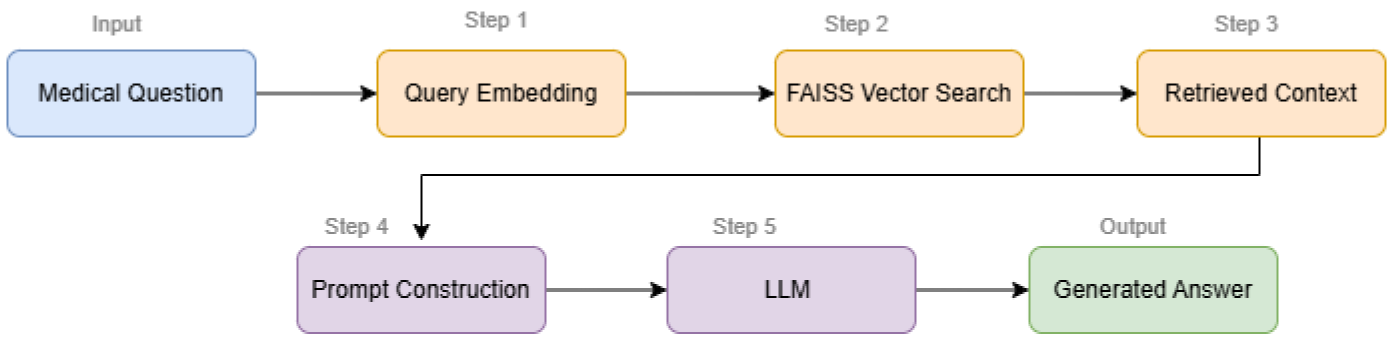

Figure 7.2: Naive RAG Architecture

7.6.2 Sparse RAG Architecture

Sparse RAG replaces semantic retrieval entirely with keyword-based retrieval using the BM25 algorithm. As shown in Figure 7.3, the medical question is tokenised into query terms which are matched against the raw textbook chunks using BM25 term frequency ranking. The top-k matching chunks are retrieved and passed to LLM for answer generation. The BM25 scoring function is:

$$BM25(q,d) = \sum_{t \in q} IDF(t) \cdot \frac{f(t,d)(k_1 + 1)}{f(t,d) + k_1 \left(1 - b + b \frac{|d|}{avgdl}\right)}$$

Where, q represents the query and d refers to a document or text chunk, t denotes a term in the query and f(t, d) is the frequency of that term in document d. The length of a document is represented by |d|, while avgdl refers to the average document length in the corpus. The parameter k1 controls how much-repeated terms influence the score (typically between 1.2 and 2.0), and b is used for document length normalization. IDF(t) represents the inverse document frequency of a term.

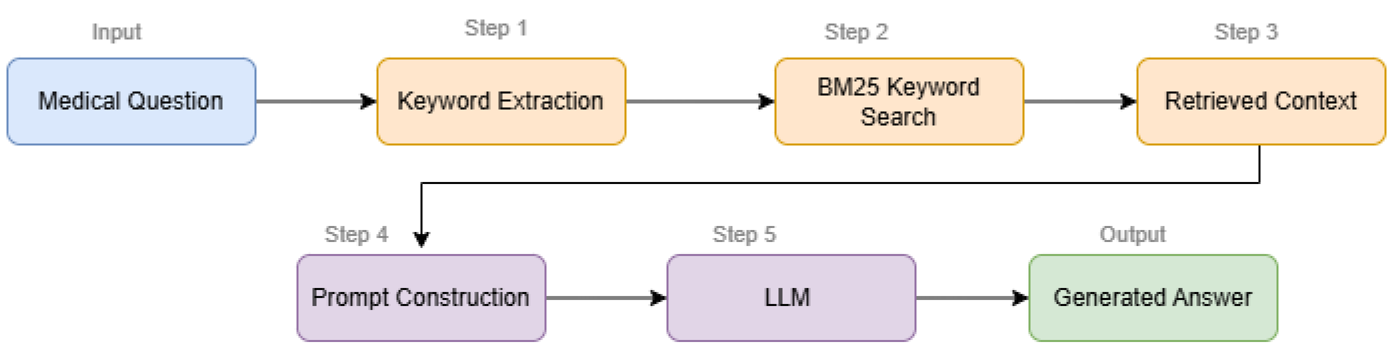

Figure 7.3: Sparse RAG Architecture

7.6.3 Hybrid RAG Architecture

Hybrid RAG combines both retrieval methods to improve accuracy and coverage. As shown in Figure 7.4, the medical question is processed through two parallel paths. The semantic path retrieves chunks via FAISS cosine similarity while the keyword path retrieves chunks via BM25 term frequency ranking. The results from both paths are merged using Reciprocal Rank Fusion (RRF, k = 60) and the fused context is passed to LLM for answer generation. Reciprocal Rank Fusion, computes a combined score using the formula:

$$RRF(d) = \sum_{r \in R} \frac{1}{k + rank_r(d)}$$

Where d refers to a retrieved document or text chunk, R represents the set of retrieval methods used and rank_r(d) denotes the rank position of document d in a given retrieval method r. The parameter k=60 is a constant.

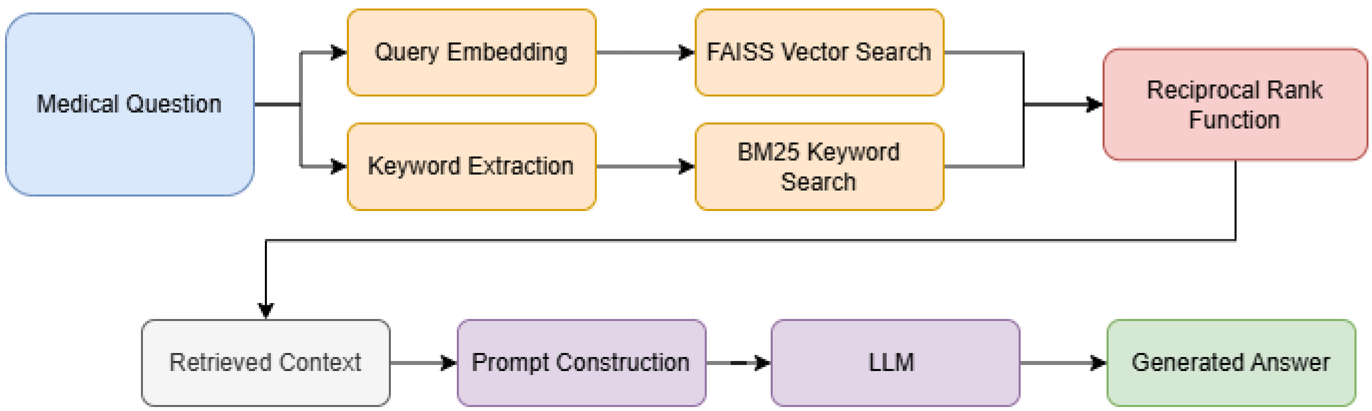

Figure 7.4: Hybrid RAG Architecture

7.6.4 Multi-Hop Explainable RAG Architecture

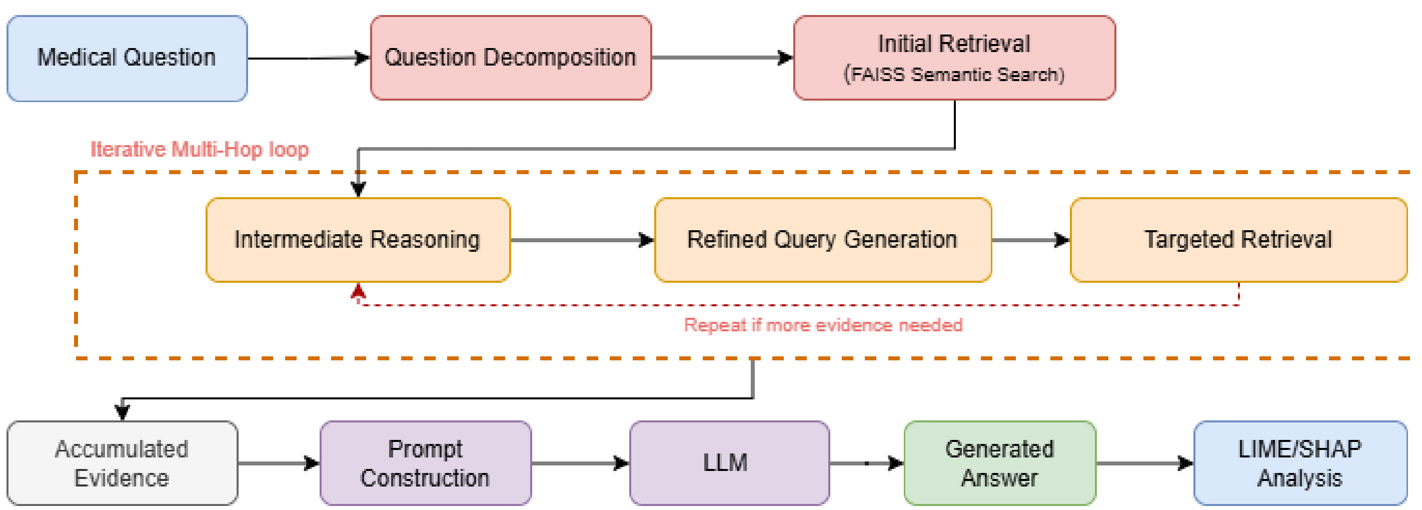

Figure 7.5: Multi-Hop RAG Architecture

The system is configured with a maximum of 3 hops per question. Hop 1 retrieves initial evidence for each decomposed sub-question using FAISS semantic search. Hop 2 analyses the Hop 1 evidence, identifies gaps and generates refined queries for targeted retrieval of missing information. Hop 3 performs a final targeted retrieval to fill any remaining evidence gaps before answer generation. The loop terminates when all sub-questions have supported evidence or the maximum hop count is reached. Question decomposition is handled by the LLM which breaks the original medical question into focused sub-questions each targeting a specific piece of clinical information needed to answer the full question. After answer generation LIME and SHAP are applied as a post-hoc evaluation step at the passage level to trace which evidence from each hop influenced the final answer.

NOTE: Final Prompting technique, LLM, Embedding and Chunking strategy will be selected based on the time availability and evaluation.

7.7 Hallucination Detection and Control Mechanism

When a language model generates confident sounding statements that are not supported by the provided evidence it's called hallucination. It is one of the most critical risks in medical AI.

7.7.1 Hallucination Detection Using RAGAS Metrics

The primary method for detecting hallucination is the Faithfulness metric from the RAGAS evaluation framework(Es et al., 2025). Faithfulness measures whether the claims made in the generated answer are actually supported by the retrieved context. Additional hallucination indicators include Answer Correctness, comparing the generated answer against the known correct answer using both F1-based factual overlap and semantic similarity and low Context Recall scores indicating that the retrieval step failed to find sufficient relevant evidence which increases the likelihood of the model filling gaps with fabricated information.

7.7.2 Hallucination Reduction Strategy

Hallucination is controlled through three layers applied at different stages of the pipeline. This layered approach ensures that hallucination is prevented where possible, detected when it occurs and flagged before reaching the end user.

Table 7.7: Three-Layer Hallucination Control Strategy

Layer	Strategy	How It Works
Layer 1	Evidence-Grounded Prompt Design	Prompt explicitly instructs the model to answer based only on retrieved evidence and to flag when evidence is insufficient rather than guessing
Layer 2a	Retrieval Optimization	Comparing four retrieval architectures identifies which approach retrieves the most complete and relevant evidence, reducing information gaps
Layer 2b	Multi-Hop Retrieval Validation	Each retrieval step is validated before proceeding to the next, creating checkpoints that catch unsupported claims early in the reasoning chain
Layer 3	Answer Rejection Mechanism	When the RAGAS Faithfulness score falls below 0.5, the system flags the answer as potentially unreliable rather than presenting it as confident output

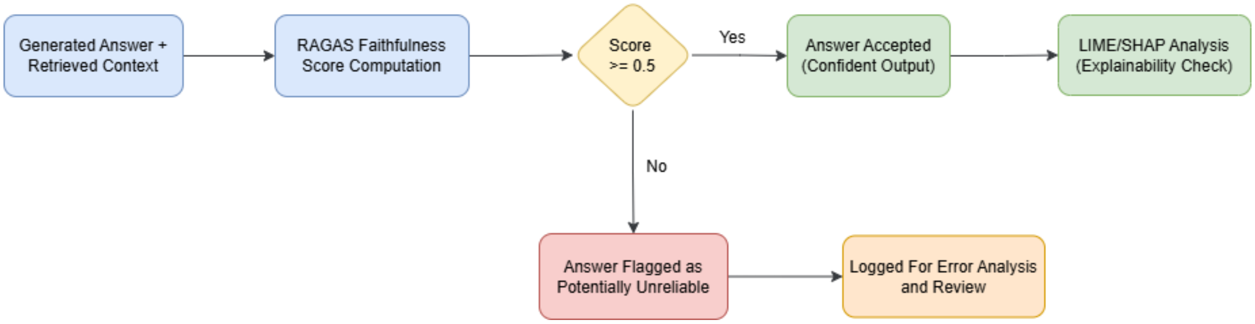

Figure 7.6: Answer Acceptance Decision Flow

7.8 Explainability Analysis Using Explainable AI Techniques

In medical AI, it is not enough for a system to give the right answer. Clinicians need to understand why a particular answer was generated and which pieces of evidence supported it. This section describes how explainability will be evaluated using two established XAI methods LIME and SHAP.

7.8.1 Explainability Using LIME

Local Interpretable Model-agnostic Explanations introduced by (Ribeiro et al., 2016). It generates local explanations by changing input features and observing how the output changes. For this research, LIME is adapted to work at the passage level. The output is a ranked list showing which retrieved passages had the most influence on the generated answer. This directly supports clinical explainability as a doctor can see which textbook passages the system relied on to reach its conclusion.

7.8.2 Explainability Using SHAP

SHAP was introduced by (Lundberg & Lee, 2017). It computes Shapley values representing each feature's marginal contribution across all possible feature combinations. SHAP provides stronger mathematical guarantees than LIME including consistency and local accuracy. For this research, SHAP is applied at the passage level where each retrieved chunk is treated as a feature.

Table 7.8: Comparison of XAI Methods Used

Feature	LIME	SHAP
Approach	simplified local model used to explain single prediction	Shapley value computation
Scope	Local explanations per query	Local and global explanations
Computation Cost	Moderate	Higher

7.9 Evaluation Framework

The evaluation framework combines two types of metrics, LLM-based metrics from the RAGAS framework and non-LLM metrics that provide objective, deterministic measurements independent of any language model evaluator.

7.9.1 RAGAS Evaluation Metrics

The RAGAS framework provides automated and reference-based evaluation specifically designed for RAG systems. This research uses five core RAGAS metrics each targeting a specific aspect of RAG performance:

Table 7.9: RAGAS Evaluation Metrics

Metric	What It Measures	Formula
Faithfulness	Whether generated claims are supported by retrieved context	$\frac{1}{M}\sum_{i=1}^{M} \mathbb{1}[\text{claim}_i \text{ is supported by context}]$
Answer Correctness	Overall correctness of the answer against ground truth	$0.75 \cdot F1(\text{TP,FP,FN}) + 0.25 \cdot \cos(\mathbf{a}, \mathbf{g})$
Context Precision	Whether relevant chunks are ranked above irrelevant ones	$\frac{1}{K}\sum_{k=1}^{K} \text{Precision@k} \cdot \text{Rel}_k$
Context Recall	Whether retrieved context contains all necessary information	$\frac{1}{G}\sum_{j=1}^{G} \mathbb{1}[\text{gt\_claim}_j \in \text{context}]$
Answer Relevancy	Whether the answer actually addresses the question asked	$\frac{1}{N}\sum_{i=1}^{N} \cos(\mathbf{q_i}, \mathbf{q})$

7.9.3 Non LLM Evaluation Metrics

Since RAGAS relies on an LLM as evaluator which can introduce phrasing bias, two additional non LLM metrics are included to provide objective and deterministic evaluation.

1. Exact Match Accuracy performs a simple string comparison between the predicted option (A, B, C or D) and the ground truth answer, reported as a percentage across all 12723 questions per architecture.
2. Retrieval Recall measures the proportion of questions where at least one of the top-k retrieved chunks contains information relevant to the correct answer, helping distinguish whether poor answers are caused by retrieval failure or generation failure.

7.10 Experimental Workflow and Execution

Figure 7.7 presents the end-to-end experimental workflow for this study. The pipeline begins with the two data sources, the MedQA dataset and the textbook corpus. It flows through data cleaning, text chunking, embedding generation and vector store indexing. The prepared data is then fed into all four RAG architectures which share a common LLM. This produces 50,892 generated answers across 12,723 questions and 4 architectures. The outputs are evaluated through three parallel assessment streams covering RAGAS metrics, explainability analysis and hallucination detection. These results converge into a comparative analysis with statistical testing to identify the optimal RAG framework for medical question answering.

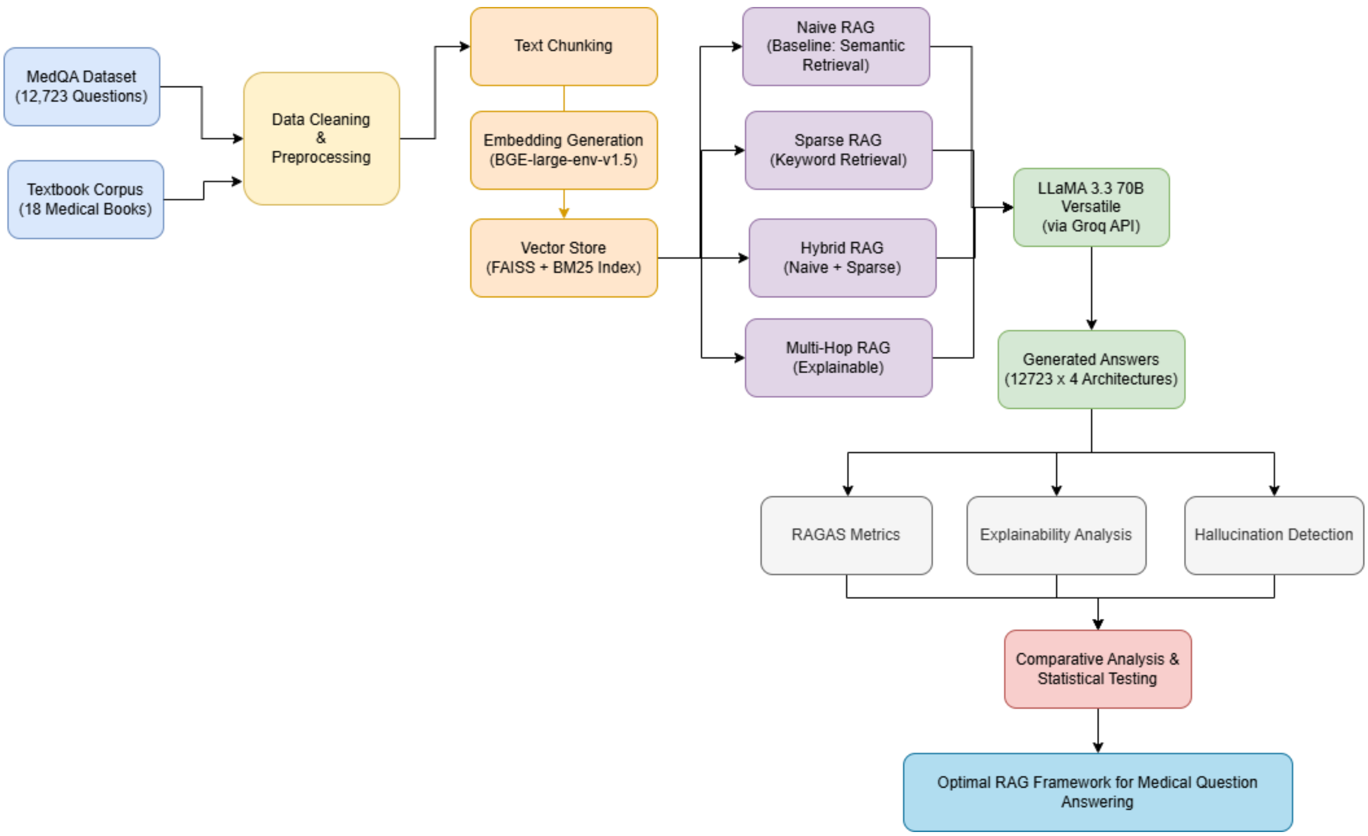

Figure 7.7: Experimental Workflow Pipeline

7.11 Ethical Considerations and Limitations

7.11.1 Ethical Considerations

This research follows ethical standards for AI research in healthcare. Although the study involves medical question answering, it does not interact with real patients or use any private clinical data. The following measures ensure ethical compliance:

- The MedQA dataset is publicly available under a CC-BY-4.0 license and contains no personally identifiable patient information.

- The hallucination detection mechanism (Section 7.7) flags low-confidence answers rather than presenting potentially incorrect medical information as fact.
- The system is a research tool only and is not intended for clinical use without further validation, regulatory approval and governance integration.
- All experiments use open-source tools, publicly available datasets and documented parameters to ensure full reproducibility.

7.11.2 Limitations of the Study

Like any research project, this study has limitations that should be acknowledged upfront:

- MedQA covers only USMLE-style questions and may not generalize to clinical notes, patient conversations or real-time diagnostics.
- Results are specific to LLaMA 3.3 70B and may vary with different language models.
- The knowledge base is limited to 18 medical textbooks and excludes clinical guidelines, case reports and recent literature.

7.12 Methodology Summary

A comparative experimental approach is followed to evaluate four RAG architectures, Naive RAG, Sparse RAG, Hybrid RAG and Multi-Hop Explainable RAG all tested on the MedQA USMLE dataset containing 12,723 clinical questions. All architectures share the same LLM backend, prompt template and FAISS/BM25 retrieval infrastructure built from 18 medical textbooks ensuring that retrieval strategy is the only variable under comparison. Evaluation covers three layers: factual accuracy against known correct answers, hallucination and retrieval quality through five RAGAS metrics and explainability analysis using LIME and SHAP. Statistical significance testing and ablation studies support the comparison with the final outcome being a deployment-oriented framework that maps clinical requirements to the most suitable RAG architecture.

8. Resource Requirements

8.1 Hardware Resources

All local experiments including data preprocessing, chunk generation, vector database operations and evaluation will be conducted on a laptop with the below configuration:

- Operating System: macOS
- System: MacBook Pro with Apple M1 Pro chip
- Memory: 16 GB Unified Memory
- Storage: 1 TB SSD
- GPU: Not required (LLM inference is handled via Groq Cloud API)

8.2 Software Resources

Table 8.1: Software Resources

Category	Tools and Libraries	Purpose
Development Environment	Google Colab	Environment management, interactive development and experiment runs
Programming and Data Handling	Python 3.13, Pandas 2.2.4, NumPy 2.3.2	Data preprocessing, chunk management and results analysis
LLM and Prompt Management	LangChain, Sentence-Transformers	Prompt flows, retrieval chains, LLM calls and dense embedding generation
Retrieval and Vector Store	FAISS, rank-bm25	Dense embedding storage and similarity search, sparse keyword-based retrieval
Model Hosting and Inference	LLaMA 3.3 70B via Groq Cloud API	Answer generation across all four RAG pipelines
Evaluation and Explainability	RAGAS, LIME, SHAP	RAG evaluation metrics computation and explainability analysis
Visualisation	Matplotlib 3.10.5, Seaborn 0.13.2	Plots and comparison charts for the final report
Version Control	Git, GitHub	Code versioning and repository management

8.3 Dataset Requirements

The MedQA dataset will be used as the primary benchmark. It contains 12,723 USMLE-style clinical multiple-choice questions with verified correct answers. The retrieval knowledge base consists of 18 medical textbooks covering anatomy, pathology, pharmacology and other clinical subjects. The dataset is publicly available, contains no patient data and is open access for research purposes. Data leakage will be prevented by ensuring that test questions and their exact contexts are not included in the retrieval corpus.

9 Research Plan

9.1 Gantt Chart

The Gantt chart presents the planned weekly schedule from March to June 2026 covering all major phases of the research.

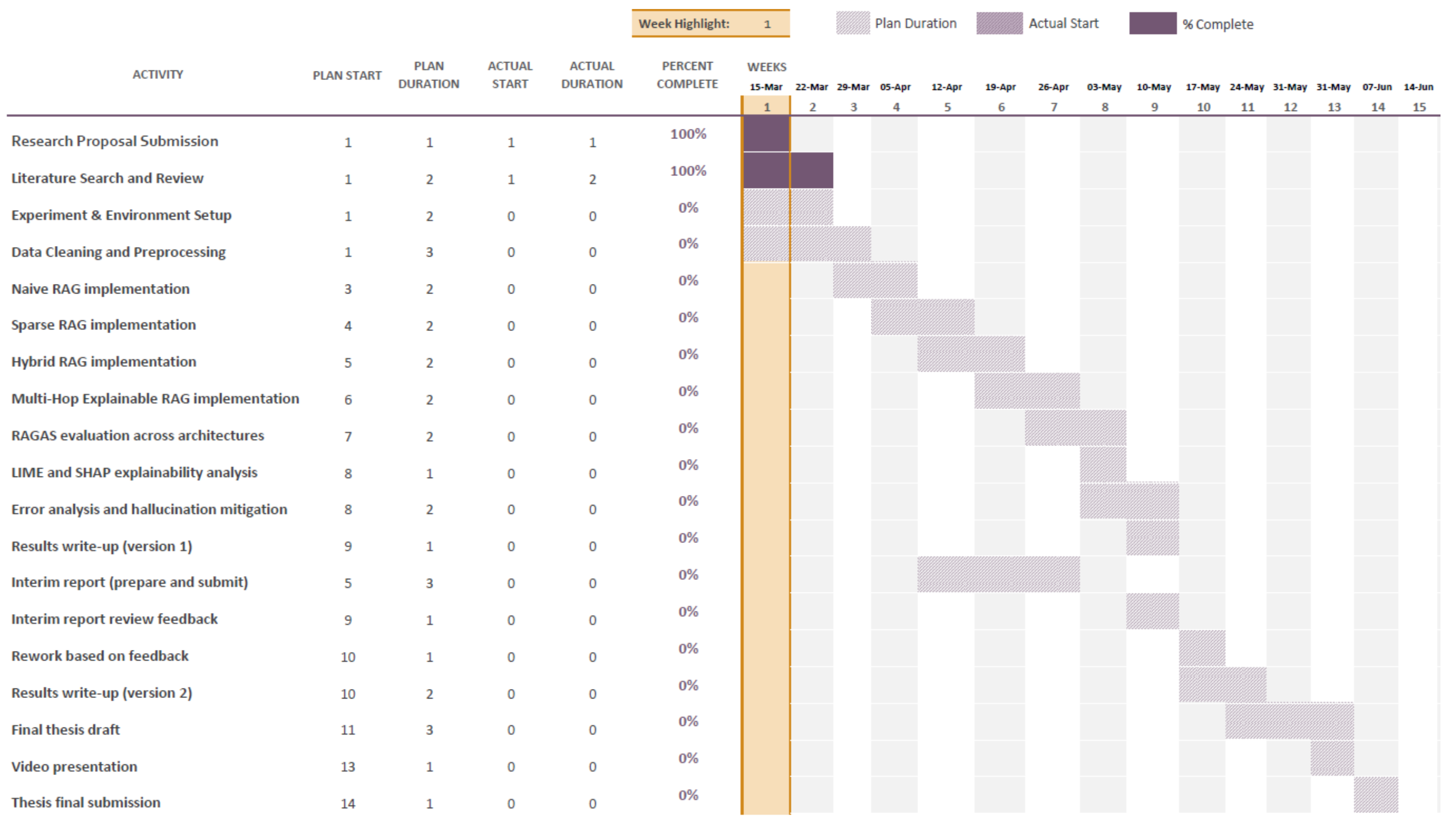

Figure 9.1: Research Gantt Chart

9.2 Risk and Mitigation Plan

Every research project carries potential risks that could delay progress or affect the quality of outcomes. The table below identifies the key risks for this study and outlines practical mitigation strategies for each.

Table 9.1: Risk and Mitigation

Risk	Probability	Mitigation Strategy
Groq API downtime or rate limiting	Medium	Switch to Together AI or Hugging Face Inference API using Qwen 2.5 72B as a backup model
Poor retrieval quality from vector database	Medium	Test multiple chunking strategies and tune chunk size and overlap during the validation phase
LLaMA 3.3 generating hallucinated medical answers	High	Apply evidence grounded prompting with a three-layer hallucination control strategy and a Faithfulness threshold of 0.5
Dataset contamination from pre-training overlap	Medium	Compare LLM performance with and without retrieval to detect potential memorisation effects
Hardware failure or data loss	Low	Maintain regular backups on Google Drive and GitHub with version-controlled experiment outputs
RAGAS evaluation producing inconsistent scores	Low	Include non LLM metrics as objective cross-checks alongside RAGAS

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